



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (EVT) (CBCS) Degree Examination
 January 2023

22EVT101 – ELECTRIC AND HYBRID VEHICLES

Duration: 3 Hours

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Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO	
1.	a) State and explain the key Electric Vehicle technologies.	6	L*2	1	3,
	b) Write a note on the followings: i) Fixed gearing and ii) Variable gearing.	6	L2	1	3,4
	c) Write a note on force parameters applicable to the EV system. Also draw the typical Force- Speed characteristics of (i) ICEV with variable gearing and (ii) EV with fixed gearing.	8	L3	1	3,4
2.	a) Obtain the comparison between EV and ICEV. Also draw the (i) Mechanical differential and (ii) Electronic differential diagram applicable to EV.	10	L3	1	3,4
	b) Draw the neat sketches of EV configurations due to the variations in energy sources system.	10	L3	1	3,4
Unit – II					
3.	a) What are the main requirements of EV motors? Also obtain the classification of commutatorless motors applicable for EV electric propulsion.	10	L2	2	3,4
	b) State and explain the requirements of preferred power devices for EV propulsion. Also, with a neat circuit diagram explain voltage source inverter fed electric propulsion system.	10	L4	2	3,4
4.	a) With a neat circuit diagram explain four quadrant DC chopper applicable for EV propulsion system.	8	L4	2	3,4
	b) With a neat circuit diagram explain the topology of three phase voltage fed resonant dc link inverter fed electric propulsion system.	8	L4	2	3,4
	c) Write a note on PM hybrid.	4	L3	2	3,4
Unit – III					
5.	a) With a neat circuit diagram and wave forms explain two quadrant DC controlled chopper applicable to EV drive.	10	L4	3	3,4
	b) With relevant wave forms explain the Equal-area PWM of EV induction motor drive system.	10	L4	3	3,4
6.	a) With a neat circuit diagram, equivalent circuit and wave forms explain two-quadrant zero-voltage transition converter-fed dc motor drive in regenerating mode.	10	L4	3	3,4
	b) With a neat block diagram explain the Field oriented control (FOC) of induction motor drives.	10	L4	3	3,4

Unit – IV

With a neat functional block diagram explain the following HEV configurations:-

- i. Parallel hybrid
- ii. Complex hybrid

10 L4 4 3

With neat functional block diagrams explain operating modes of electric-heavy series-parallel hybrid HEV configuration.

10 L4 4 3

With neat functional block diagrams explain operating modes of Series hybrid operating HEV configuration

8 L4 4 3

Mention the key considerations for design of power control strategies for HEVs. Also with help of neat functional block diagrams explain operating modes of Parallel hybrid operating HEV configuration.

12 L4 4 3

Unit – V

9. a) Obtain the classification of EV batteries. Also Explain the following EV infrastructures: (i) Normal charging infrastructure, and (ii) Fast charging infrastructure.

12 L4 5 3

- b) With a neat diagram explain the Soft-switching power converter for inductive charging applicable to EVs.

8 L4 5 3

10. a) Write an explanatory note on the following EV batteries:-

- i. Sodium- β based batteries.
- ii. Ambient - temperature lithium based batteries.

6 L3 5 3

- b) Draw the charging characteristics (cell voltage vs. input capacity) of various batteries.

6 L4 5 3

- c) With a neat diagram explain the basic principle of ultra capacitors.

8 L4 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
